F220 Operator Manual

1. INTRODUCTION

The Pertronic F220 is a modular, expandable, analogue addressable automatic fire alarm system designed for medium to large building applications. It is expandable from 2 to 20 addressable loops giving a maximum of up to 3180 addressable smoke or heat detectors plus up to 1980 manual call points, modules, or addressable relays for the system.

The panel uses a 7 inch (180 mm) 800 x 480 pixel colour display to unmistakably identify the panel status. The alarm mode is clearly identified by the use of red status bars, and by using large easy to read text descriptions. Fault information, device disablement information, pre-alarm conditions, walk test, supervisory and system information all have their own unique coloured display screens to provide comprehensive easy to use information for all users including fire brigade personnel, building managers, building owners and service technicians.

A lockable front door provides access to the Keyboard-Display.

The **F220** conforms to Australian Standards AS 7240.2–2004, AS 7240.4–2004 and AS 4428.3–2010.

This Operator Manual is intended for building owners or building managers. It does no describe all functions and system features nor does it cover all menu items. For more information contact your fire alarm system service company.

For building owners/managers, Pertronic Firemap® is recommended for fire alarm system management. Contact your fire alarm system service company for more information.

Figure 1.1 Pertronic F220 Analogue Addressable Fire Alarm Control Panel (16U Cabinet)

2. ANALOGUE ADDRESSABLE LOOP

Detection circuits in the Pertronic F220 fire alarm system use analogue addressable loop architecture. A system may have up to 20 analogue addressable loops. Each loop consists of a single-pair cable connected in a loop, from the loop driver (**A** terminals), to each analogue addressable device in turn, and finally back to the loop driver, (**B** terminals). The loop is effectively driven from both ends. This means that an open circuit fault, anywhere on the loop, will not affect its operation.

Analogue addressable fire system devices include smoke detectors, heat detectors, manual call points, monitor modules, loop responders, and loop relay units. A mixture of device types may be used.

Each loop provides detector addresses 1 to 159 for smoke and heat detectors, and module addresses 1 to 99 for manual call points, relays, loop responders, etc.

Up to 258 devices may be installed on each loop (159 detectors plus 99 modules). Each detector or module must have a unique address. No two detectors can have the same address and similarly no two modules can have the same address.

The zones within a loop are determined by the premises layout and the routing of the loop wiring. Each zone is allocated a zone number and can be allocated a 31 character zone description.

The F220 detects loop faults by performing a continuity test before each loop poll sequence. Loop isolators are used between zones to prevent more than one zone being lost in the event of a loop short-circuit.

Figure 2.1 illustrates a system with two analogue addressable loops, showing detectors (D), modules (M), isolators (ISO) and a typical device labelling scheme.

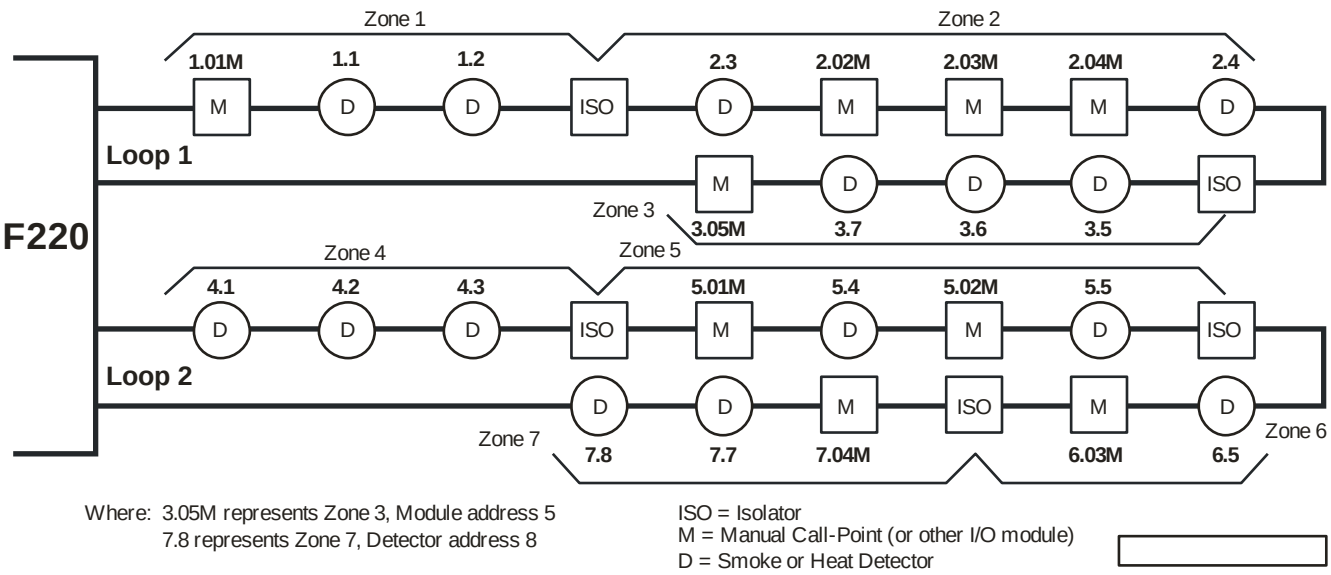


Figure 2.1: Typical Device Labelling

3. F220 FRONT PANEL

The front panel of the **F220** contains an 800 x 480 pixel colour alphanumeric LCD display.

The display is within the **Fire Brigade Panel** (enclosed by a red border), together with push-button controls and LED indicators.

Outside the **Fire Brigade Panel**, additional engineering push-button controls and LED indicators are available for use by the building manager and fire alarm engineers.

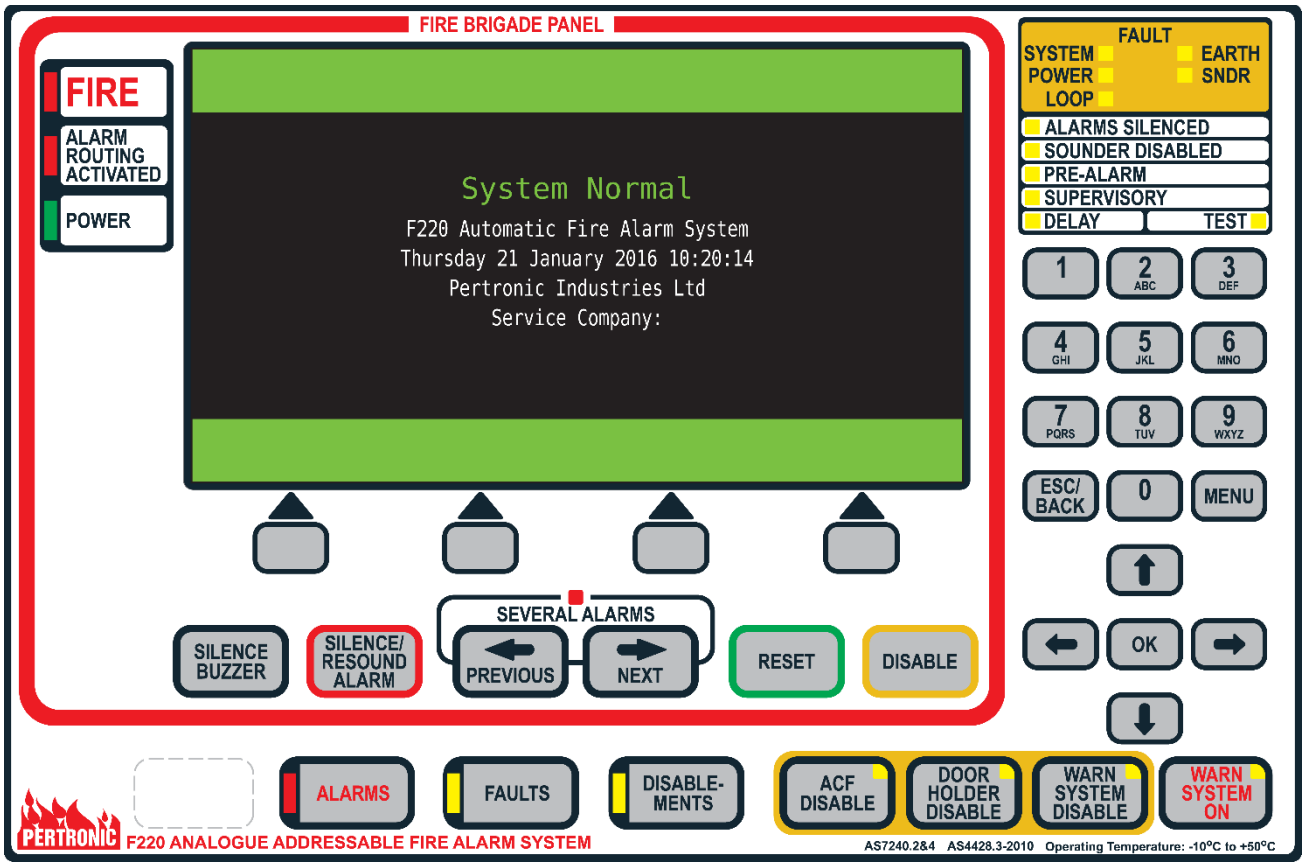


Figure 3.1: F220 Front Panel

4. ALPHANUMERIC LCD DISPLAY

4.1 Event Display and Event Queues

Whenever an event occurs on the F220 system it is placed into one of the queues in the table below. The queues are colour coded on the display.

Queue		Contains / example	Priority
Fire	**Fire**	Devices in Alarm	Highest
Walk Test	**Walk Test**	Zones that have “Walk Test” alarms	
Pre-Alarm	**Pre-Alarm**	Detectors that are very near the alarm threshold	
Fault	**Fault**	Contains all panel faults such as “double address”, missing devices, power supply faults, system faults, loop faults earth faults, sounder faults	
Disablement	**Disablement**	All devices and zones that have been disabled	
Supervisory	**Supervisory**	Outputs of systems monitoring suppression systems , air handling systems or other life, safety or property protection systems.	
System Events	**System Events**	An event which is not related to the function of the system, that affects the system's ability to perform it's function.	
Active Events	**Active Events**	Typically inputs from monitoring devices that are “off normal”.	Lowest
System Normal	**System Normal**	Normal View (no events in any queue).	

An event queue that contains an event is deemed active and the event queue with the highest priority will be displayed. If other queues are active with lower priority events this will be indicated by the appropriate LED indicator and the user can view these if required.

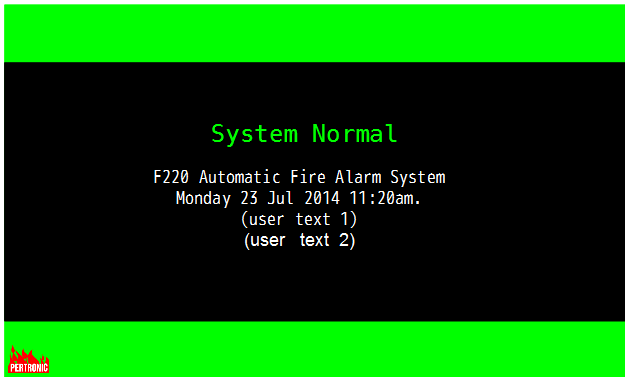
The user may switch between active queue displays by repeatedly pressing the **OK** button.

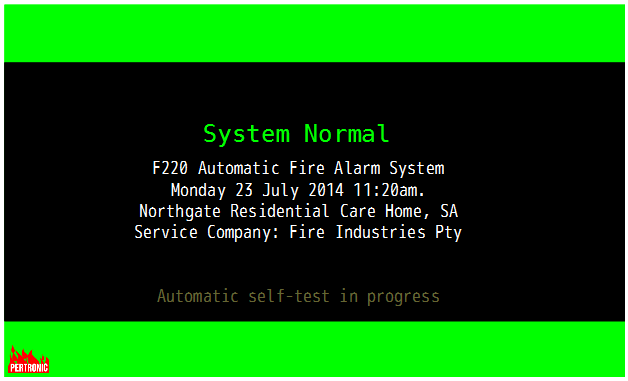
The “Fire” queue contains devices in alarm. If all the alarms have been either reset or disabled the “Fire” queue becomes inactive.

4.2 Normal / Date-Time Display

When the F220 is Normal the following screen is displayed showing date / time and two lines of user text. A variant of the Normal screen is also shown when the system is performing background testing.

These displays may also appear after 10 minutes of no keypad activity.

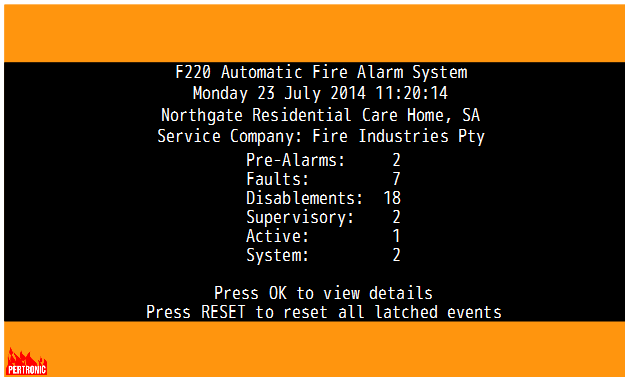




4.3 System Off-Normal Display

This display will occur if activity is present in the system that is not an alarm or the system is not in Walk test. The colour of the band at the top and bottom of this Off-Normal summary display will be that of the highest priority event being reported.

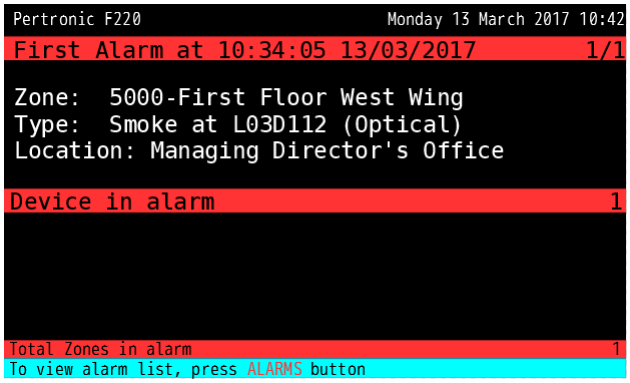
The example shown here has “orange” pre-Alarm bands showing.

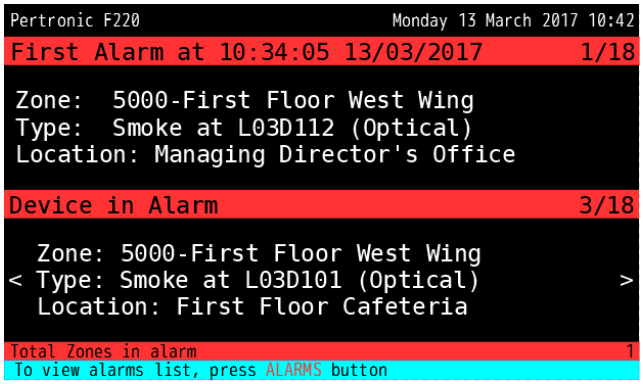


4.4 Fire Alarm Display

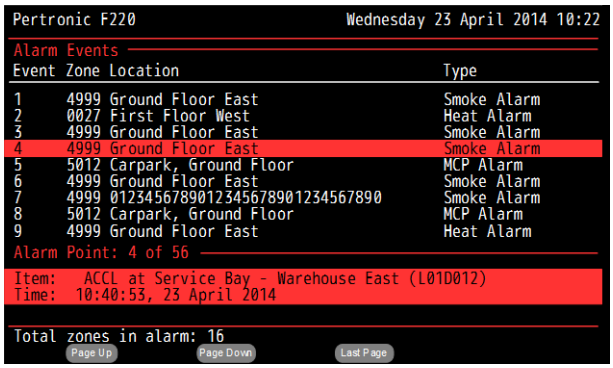
The First Alarm present on the CIE is always presented at the top of the display and remains until the system has been reset.

Any subsequent alarms that occur are displayed below first alarm and multiple subsequent alarms can be displayed using Previous or Next keys of the Several Alarms function.





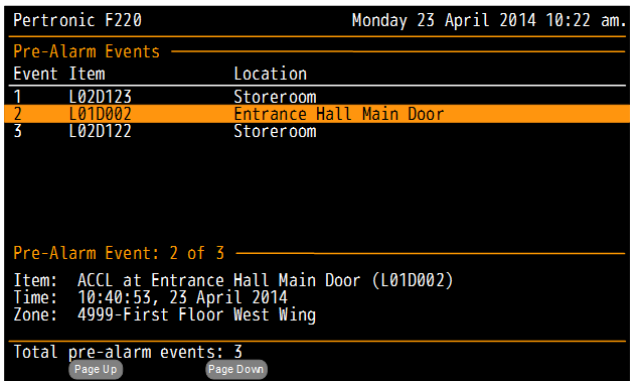
Whenever alarms are present the “Alarms” LED will be ON indicating that users can press the associated key to view details of all points that currently are in alarm (“points” includes: devices, logic blocks, timers etc.).



In the example shown there are a total of 56 points in alarm (“points” includes: devices, logic blocks, timers etc.) and these can be viewed by using the Page Up or Page Down soft keys. Select a particular event using the keyboard ↑ or ↓ keys.

4.5 Pre-Alarm List Display

The Pre-Alarm list will be displayed if it is the highest priority active queue. If alarms are present as well these will have priority and the presence of Pre-Alarms is indicated by the Pre-Alarm LED turning **ON**—press the **OK** key to view the Pre-Alarm list.



Pre-Alarm events can be selected using the keyboard ↑ or ↓ keys and once selected can be **Reset** or **Disabled**

Use Page Up or Page Down soft keys to view information not currently on-screen.

5. FRONT PANEL INDICATORS AND CONTROLS

5.1 Fire Brigade Panel (AS 4428.3-2010)

The red border as shown in Figure 3.1: F220 Front Panel encloses the Fire Brigade Panel.

5.1.1 Fire Brigade Panel Indicators

Indicator	Function	Colour	Description
	FIRE	Red	General **Fire Alarm** Indicator.
	Alarm Routing Activated	Red	Indicator for transmitted alarm routing equipment activated. Remains **ON** until the **fire alarm** condition is reset.
	Power	Green	**ON** when mains or battery power is available.
	Several Alarms	Red	Indicator for several alarms. **ON** if three or more devices are in alarm.

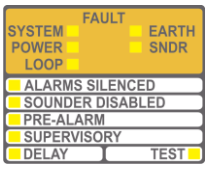
5.1.2 Fire Brigade Panel Controls

Control	Function
Silence Buzzer	Press to Silence the CIE alarm / fault buzzer. Note: Buzzer resounds for new Alarms
Silence / Resound Alarm	Press to Silence sounder devices or Resound alarm devices if the “Alarms Silenced” indicator is **ON**—toggle action.
Previous, Next (Several Alarms)	Scroll keys used to view devices in alarm. The scroll keys will only have effect if if three or more devices are in alarm.
Reset	Press to Reset the fire alarm panel to its Normal condition. Any faults remaining after the CIE has reset will be re-established within 20 seconds.
Disable	Press to disable the zone in Alarm. Also can be used to return devices that were “disabled” to their Normal state.

5.2 Engineering Indicators and Controls

The indicators and controls outside the Fire Brigade Panel are for the use of the building owner, manager or service and maintenance personnel. These are shown in Figure 3.1: F220 Front Panel.

5.2.1 Engineering Indicators

Function	Colour	Description
Alarm	Red	Indicates that devices are in alarm and can be viewed by pressing the associated key.
Fault (Common Fault)	Yellow	Common fault indicator. ON whenever any fault is present on the system. In addition one or more fault category indicators will also be ON.
Disablements	Yellow	Indicates that devices are disabled. Disablements can be viewed on the display pressing the associated key.
		
System	Yellow	**ON** if the internal system software is not running properly
Earth	Yellow	**ON** when an earth fault is detected by the CIE
Power	Yellow	**ON** when there is a power fault with the CIE or devices connected to the loop eg Mains missing, battery test failure
Sndr (Sounder)	Yellow	**ON** when a sounder fault is present eg wiring open or short
Loop	Yellow	**ON** when a fault is related to a device on the loop or loop wiring (open, short, isolator…)
Alarms Silenced	Yellow	**ON** if alarm devices have been silence by operation of “Silence / Resound Alarm”
Sounder Disabled	Yellow	**ON** if Warning System devices have been disabled
Pre-Alarm	Yellow	**ON** when detectors have gone off-normal and potentially could soon go into an Alarm state
Supervisory	Yellow	**ON** when a supervisory input is in the active state.
Delay	Yellow	**ON** when one or more output circuits are in a delayed operating mode
Test	Yellow	**ON** when one or more zones are in Walk Test mode

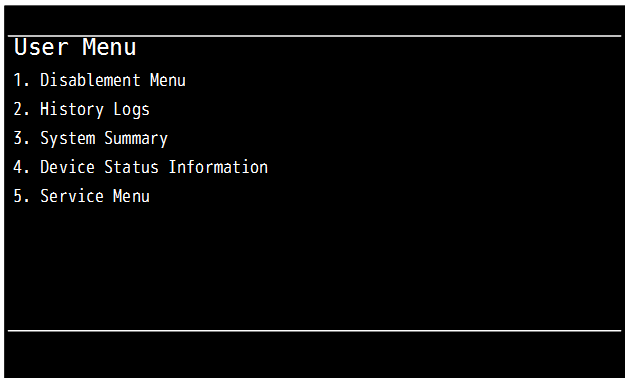
5.2.2 Engineering Controls

Control	Function
Alarms	Press to view a list of all points that are currently in Alarm state (“points” includes: devices, logic blocks, timers etc.)
Faults	Press to view a list of all current fault events
Disablements	Press to view a list of all devices currently disabled
ACF Disable	Press to toggle the disable state of **ACF** type devices (such as fan controllers or other equipment)
Door Holder Disable	Press to toggle the disable state of **Door Holder** type devices (prevents door release)
Warn System Disable	Press to toggle the disable state of **Warning System**
Warn System On	Press to turn **ON** or **OFF** the building **Warning System** (manual evacuation)
Function Keys	These four keys select the soft-key functions displayed at the bottom of the display (often ‘page-up’ and ‘page-down’)
OK key	Used to select a highlighted menu item.
Menu key	Used to open and exit User Menus.
Navigation Keys	Used to move between “fields” while viewing lists or “editing” functions. Can also be used to move a highlight in menus.
Numeric Keys	Can be used to make a menu selection and in some menus when a numeric value is required e.g. entering a zone number

6. USER MENU OPERATIONS

The User Menu is used for accessing functions such as Disablement functions, History Logs, System Summary, Device Status Information and Service Menu. Not all menu items are described but the user interface for most user menu items is similar.

The First Level User Menu is reached by pressing the **Menu Key** followed by the **numeric key** corresponding to the sub-menu option required.



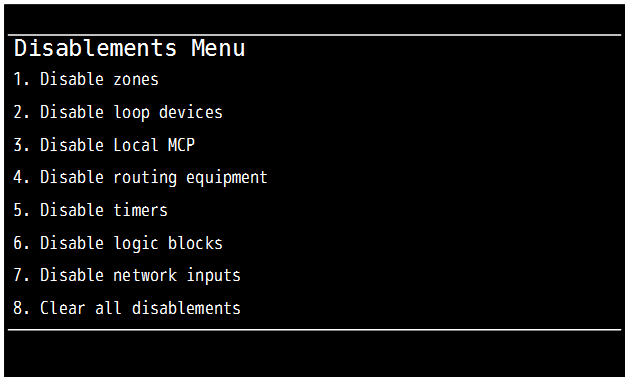
6.1 User Menu-Disablements menu (Option 1)

Press the corresponding numeric key to select the Disablement type required.

Disablements will leave parts of the alarm system inoperative.

Disablements should only be made by those authorised to do so and should be removed as soon as is practicable.

Disablements are useful for faulty sensors that are causing false alarms or where building maintenance activity, such as welding, may cause a false alarm. In either case, full function must be restored as soon as is practicable.



6.1.1 Disable Loop devices (Option 2)

Move between Loop and Detector/Module fields using the left / right arrow key.

Enter loop number either by using up/down arrow keys or numeric keypad.
Select between Detector/Module using up/down arrow keys.
Enter Detector/Module number either by using up/down arrow keys or numeric keypad.
Press the disable key to disable/enable the device.

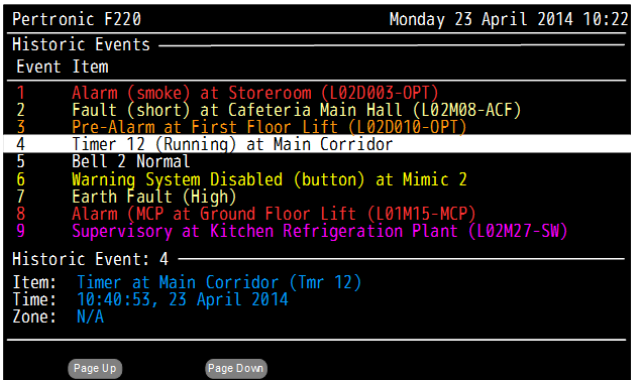
6.1.2 Clear All disablements

Pressing “8” will bring up confirmation or cancel soft-key labels as shown below.

6.2 User Menu-History Logs (Option 2)

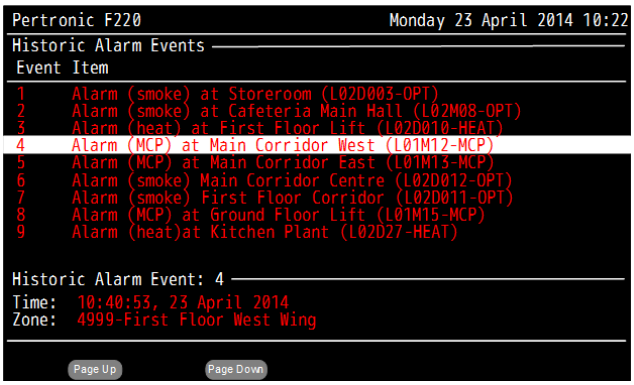
6.2.1 Main Log (Option 1)

Shows all logged events in order of occurance.



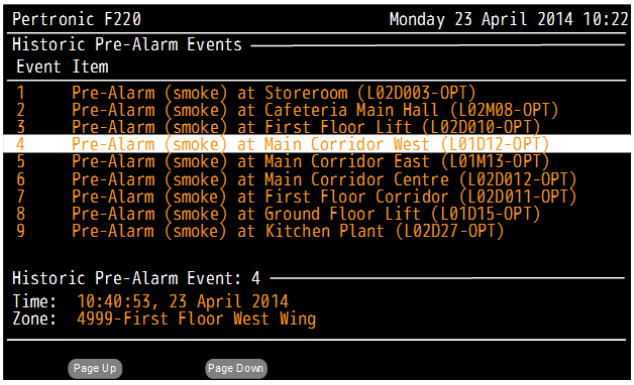
6.2.2 Alarm Log (Option 2)

Shows the Alarm log only.



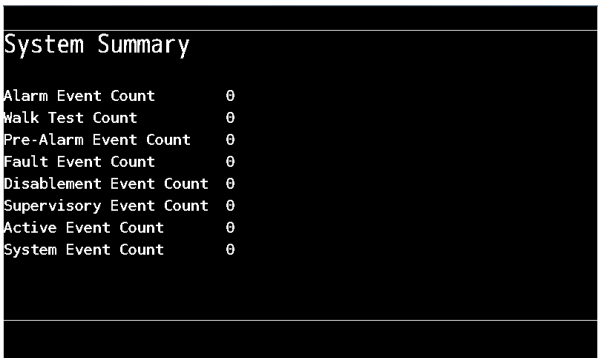
6.2.3 Pre-Alarm Log (Option 3)

Shows the Pre-Alarm log only.



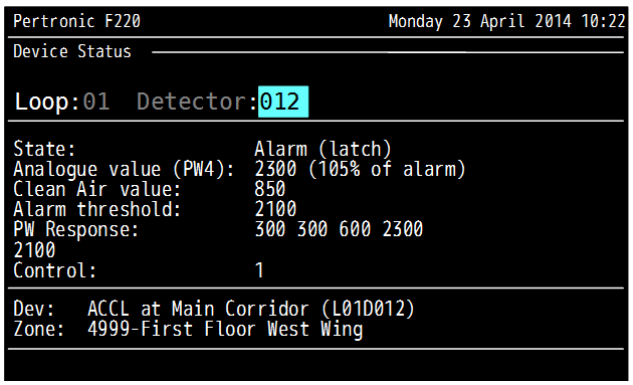
6.3 User Menu-System Summary (Option 3)

Shows a summary list of system events.



6.4 User Menu-Device Status / Test (Option 4)

Individual device status can be displayed.



7. OTHER SYSTEM FEATURES

7.1 Evacuation Test

The push-button labelled **WARN SYSTEM ON**, located on the front panel is provided to operate the External Bell and Warning System.

Operation of the **WARN SYSTEM ON** control located at mimic panels will also carry out the Evacuation Test function at the **F220**.

7.2 Buzzer (Internal Sounder)

The Buzzer is activated when any alarm or fault condition is active. The Buzzer will sound with the following cadences:

ALARM- Rapid ON / OFF

FAULT- ON continuous.

Pressing the SILENCE BUZZER key will silence the buzzer however it will re-sound if a new ALARM or FAULT condition is detected.

Pressing any push-button on the front panel (usually Enter) silences the sounder, as well as performing the action associated with that push-button.

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